

b. Starting torque:

$$T_{\text{start}} = 2i\ell Br = 2(480)(0.5)(0.5)(0.25) = 60 \text{ N}\cdot\text{m}$$

c. No-load steady-state $\rightarrow I_a = 0$

$$\therefore E_c = E - I_a R = E = 48 \text{ V}$$

$$E_c = 2vBl = 2 \times \frac{2\pi n}{60} \times Bl = \frac{r\pi n Bl}{15}$$

$$\therefore n = \frac{15 E_c}{r\pi Bl} = \frac{15 \times 48}{0.25 \times \pi \times 0.5 \times 0.5} = 3666.93 \text{ RPM}$$

d. At steady-state, $T_m = T_L = 5 \text{ N}\cdot\text{m}$

$$T_m = 2I_a \ell Br$$

$$I_a = \frac{T_m}{2\ell Br} = \frac{5}{2 \times 0.5 \times 0.5 \times 0.25} = 40 \text{ A}$$

$$E_c = E - I_a R = 48 - 40 \times 0.1 = 44 \text{ V}$$

$$E_c = 2vBl = \frac{r\pi n Bl}{15}$$

$$n = \frac{15 E_c}{r\pi Bl} = \frac{15 \times 44}{0.25 \times \pi \times 0.5 \times 0.5} = 3361.35 \text{ RPM}$$

e. The machine is a **motor**, driving a load of $5 \text{ N}\cdot\text{m}$ at 3361.35 RPM .

The speed drops from 3666.93 at no-load to 3361.35 due to increase in load.

f. Rotated by external $5 \text{ N}\cdot\text{m} \rightarrow$ rotor will speed up since external torque is added to torque that is already generated.

$n \uparrow \Rightarrow E_c \uparrow$ until $E_c > E$ and I_a changes direction,
so that machine becomes generator.

$$T_c = T_m = 5 \text{ N}\cdot\text{m}$$

$$T_c = 2 I_a l B r \quad 5$$
$$I_a = \frac{T_c}{2 l B} = \frac{5}{2 \times 0.5 \times 0.5 \times 0.25} = 40 \text{ A} \quad \text{Opposite direction from d) and e)}$$

$$E_{gm} - E = R I_a$$

$$E_{gm} = E + R I_a = 48 + 0.1(40) = 52 \text{ V}$$

$$E_{gm} = 2 v B l = \frac{r \pi n B l}{15}$$

$$n = \frac{15 E_{gm}}{r \pi B l} = \frac{15 \times 52}{0.25 \times \pi \times 0.5 \times 0.5} = 3772.51 \text{ RPM}$$

g. This machine is working as a generator.

Input torque produces mechanical power on shaft, which results in a current (and electric power) being forced into the source.